

The Potential of Quadratic Voting for Cohousing Communities

A Situated Design Case Study

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Abstract

Quadratic voting (QV) is a means of making collective decisions that include and reflect the intensity of individual voter preferences. While QV has mainly been applied to legislative case studies, it is our conjecture that QV also holds potential for decision-making processes taking place in cohousing communities. To explore this conjecture, we conducted a situated design case study in which we introduced QV into the decision-making process of the Community Land Trust H-buurt, Amsterdam (NL). Our case study includes four situated actions taken from an embedded position in the community, and yielded data ranging from screenshots, sketch notes and photos, to a QV tool, ballot issues and voting data. In this paper, we describe our case study in more detail, report and discuss our results, and present our conclusion that cohousing is ripe for QV as well as recommendations to operationalize QV in this domain.

CCS Concepts

• **Human computer interaction** > Collaborative and social computing; • **Applied computing** > Computers in other domains;

Keywords

Quadratic Voting, Situated Design, Cohousing, Decision Making, Community Land Trust

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1 Introduction

In recent years, quadratic voting (QV) has been put forth and popularized as a means of making collective decisions that include and reflect the intensity of individual voter preferences [16]. Participants in QV each receive a number of credits, which can be turned into votes, that can be allocated either in favor of or opposed to a

number of ballot issues. This mechanism allows people to express these preferences by voting with greater conviction on the ballot issues they care about most [9]. Credits convert to votes according to their square root. In this way, QV can mitigate the so-called ‘tyranny of the majority’ [7]; the domination of preferences of democratic majorities over those of minorities, disregarding the intensity of their convictions regarding an issue.

So far, QV has mainly been applied to legislative case studies to prioritize budgets [15, 20], but it has hardly been tested in real life situations. However, QV seems ripe for study in other domains in which people need to make collective decisions, especially those where it is of concern that individual preferences are expressed [2]. One of these is cohousing, a domain in which a vast number of decisions need to be made collectively on issues that typically affect individual members differently [19]. While some members of a cohousing community may, for instance, have a stronger conviction to allocate space for shared childcare, others may have a strong preference for the allocation of square meters for shared green space. It is our conjecture that QV holds potential for decision-making processes taking place in cohousing communities.

What sets cohousing apart from other QV ripe domains is that the people voting on issues share a strong social connection in addition to a common concern (cf. [19]). This is in contrast to most case studies on QV, in which both voters and issues are randomly selected, or a shared concern has been articulated but voters have no significant social relationship or simply have not met. In such cases, QV is applied to a singular coming together, such as a response to a survey [1, 14] or the one time allocation of a budget (cf. [15, 20]), while decision making in cohousing communities typically involves many gatherings and decisions. It seems that, so far, few case studies have been conducted into the application of QV for communities with long-term social stakes, that typically require longer processes of deliberation based on a previously well-defined and articulated common good, like cohousing.

Moreover, studies on QV typically evaluate and halt at the decisions made but take surprisingly little interest in evaluating the decision-making process. Illustrative for this is that such studies typically pay little attention to the design of the software tools commonly used to conduct QV. Hence, they are hardly ever evaluated in situ. Resultantly, their accessibility to novice users is limited and voting results are not used to benefit deliberation.

Therefore, to explore our conjecture that QV holds potential for cohousing communities, we conducted a situated design case study in which we introduced QV into the decision-making process of a cohousing project in Amsterdam (NL). We found that the domain

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of cohousing is ripe for QV and conclude that our situated design case study supports our conjecture. Furthermore, on the basis of our findings, we make recommendations to operationalise QV in this domain and suggest that future research on its potential should pay attention to decision-making processes, as opposed to merely evaluating a singular voting result.

In the next section, we describe our situated design case study in more detail, before reporting and discussing our results and presenting our conclusions in the subsequent sections.

2 A situated design case study: the CLT H-buurt wooncoop

The Community Land Trust (CLT) model is a way to build long-term affordable housing and ward off real estate speculation, to ensure affordable housing and neighborhood facilities in perpetuity by taking them off the real estate market. This is of special concern in the Bijlmer neighborhood of Amsterdam, where the CLT H-buurt is based. Housing prices in this neighborhood are spiking due to gentrification, on top of the ongoing housing crisis affecting the whole of Amsterdam. Anyone who supports the mission of the CLT H-buurt can become a member, by paying a yearly contribution of 10 EUR.

In 2023, the CLT began establishing their first cohousing dwelling (in Dutch also known as a ‘wooncoop’) in the Bijlmer neighborhood. The dwelling will contain 25 units, both rent controlled and for sale, to be allocated to CLT members. Supporting this endeavor are several professionals, including community organizers and an architect.

A potpourri of terms for community-based housing are used in literature, including co-living, cooperative housing, collaborative housing, community-led housing and co-housing. For the purposes of this case study, we use the term *cohousing*, which best describes a common good managed by groups that aspire to remove their project from the speculative housing market and run it under joint arrangements (cf. [19]).

2.1 Participants

The CLT H-buurt consists of 119 members at the time of the case study, 30 of them are involved in planning, organizing and designing the wooncoop. The CLT selected 15 members from their midsts for active involvement in our case study. These 15 participants ranged in ages from nineteen to members in their fifties. All of them either resided, or performed community service in the Bijlmer neighborhood at the time of our case study— both of which are indicators for wooncoop allocation. Through the ongoing process of planning, organizing and designing the wooncoop together, the CLT members (including our participants) have formed a strong social bond. Many also know each other through other neighborhood organizations, and see each other regularly at general CLT assembly meetings; and they communicate regularly in dedicated Whatsapp groups. The participants gave their informed consent to record and store any form of documentation of conversations and meetings for research purposes.

2.2 Research Method: a situated approach

As mentioned above, QV typically requires software tools to, for example, distribute credits to voters, turn credits into votes, and register votes to ballot issues. Given their importance for QV, it is surprising that such tools are difficult to access, not designed with novice users in mind and hardly ever evaluated in situ. The latter is especially problematic as it has long been established that the lived user experience and the context of use should be taken into account in human-computer interaction design [18]. This goes for design with real-world impact, such as QV, in particular [10].

In other words, designing QV, at least for real-world implementation, requires taking a situated approach. Such is a design approach in which the designer enters “into a ‘conversation’ with a situation, by a chosen methodological approach”, which “should create a dynamic in which the situation ‘talks back’, and [the designer] responds to the initial intention” (ibid.). In our case, that intention is to explore the potential of QV for decision making among cohousing communities. Although the ‘talking back’ may lead the design process away from its initial intention, as is common in situated design (ibid.).

Lave and Wenger stress that situated design processes should be embedded in communities of practice [17]. By doing so, they “highlight the interactions and interdependencies between designers, designs, design methods, and the use situation with its actors, activities, structures, particulars, and broader context” (ibid.). Situated design processes should therefore take place in particular situations, in which situated actions are carried out from embedded positions [8, 10]. Therefor, instead of exploring decision making methods and tools in the realm of cohousing in general, we purposefully focussed our case study on a specific user group and the diverse means of decision making they were trying out, by embedding ourselves in the group and carrying out situated actions from that position.

2.3 Embedded position: Becoming a CLT member

Our situated design process started by paying the yearly membership fee, which allows one to partake in general CLT assembly meetings, and explaining our interest in becoming a member of the CLT. Subsequently, we (the researchers) were invited to join two active Whatsapp groups, one for all CLT members and one dedicated to designing the wooncoop. This allowed for the observation of conversations and meeting announcements relevant to their decision-making processes. These were recorded as screenshots (see Figure 1) and stored on our research drive, with permission from the moderators.

Announcements of upcoming meetings to figure out which spaces for proposed shared facilities/activities should be included and/or prioritized were taken as prompts for direct in-person engagement with other wooncoop members through four situated actions over the period from September 2022 until July 2024.

2.4 Situated action 1: Attending meetings and design sessions

Our first situated action included attending CLT general assembly meetings. These meetings are held four times a year and are open to

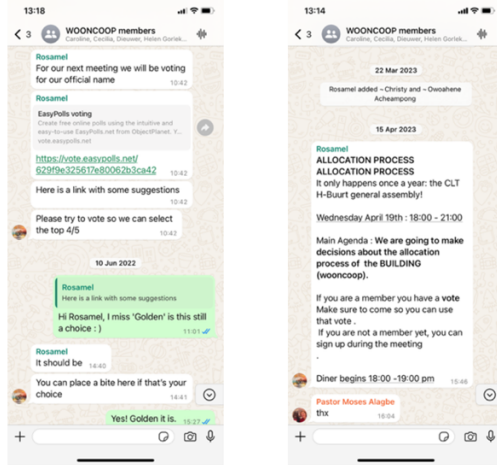


Figure 1: Screenshots of a WhatsApp group conversation (left) and a meeting announcement (right).

all CLT members. In total, we attended two CLT general assembly meetings, which both included updates about the planning, organization and design of the wooncoop. We also attended a session planned by the project architect to spatially design the wooncoop, in which she asked participants to draw their ‘dream’ dwelling. Our observations during these get togethers were documented with 3 pages of written notes and 15 photos of the activities that took place.

2.5 Situated Action 2: Developing a QV tool

Our second situated action was to design and develop a new QV tool to be used to make collective decisions about which shared spaces should be included and prioritized in the wooncoop.

After conducting preliminary tests using existing tools, it quickly became apparent that existing QV tools and interfaces were not suited to novice users. Many of the open-source QV tools consist of complex Excel sheets or formulas [4] and pages displaying results in graphs [6], which have been developed outside of a contextual user setting and are resultantly too complex to use in a real-world setting. A notable exception to this is an interactive article on QV published in the Economist [3]. Although significantly more user-friendly than most open-source QV tools, the article does not include any dissemination or analysis of voting results, and the ballot issues presented are devoid of context. Overall, the interfaces of QV tools fail to give an intuitive user overview of all ballot issues and the credit-to-vote exchange. Nor do they provide for an insightful analysis of the voting results. The results screens are especially confusing to novice users, meaning resulting voting data cannot therefore be effectively decimated, interpreted and applied in situ.

These factors led us to develop a new open-source [5] (online) QV tool [11], designed to be more suited to novice users by providing for an intuitive user overview of all ballot issues, the credit-to-vote exchange, and a more straightforward results page.

2.6 Situated action 3: Elucidating ballot issues

To elucidate issues for the QV voting ballot we organized paid (100 EUR) interviews with each of the 15 participants. We also asked them to bring physical objects with them that they would be willing to share with the other wooncoop members, to help trigger in-depth conversations about the needs and desires for different shared spaces in the wooncoop. Objects that were too large to carry, like kitchens or large screen televisions, could be photographed. This action was led by the researchers and documented in 13 pages of interview notes and a collection of 33 digital photos, including all of the objects brought to the interviews.

2.7 Situated action 4: Applying the QV tool

For our fourth situated action, using the QV tool resulting from our second situated action, we set up a voting round by inputting and sequencing the ten ballot issues resulting from our third situated action, and allocated 99 credits to each voter.

We announced in the WhatsApp wooncoop group a workshop which we led, during in which we planned to use the new QV tool to make decisions about the wooncoop’s spatial design. 14 participants attended the workshop we led at CLT headquarters in Bijlmer, Amsterdam; one participated remotely online. All voters were sent a sign-in link for the voting round. Participants used a laptop or iPad to access the online voting platform [12]. We walked them through our QV tool, including the option to remain anonymous, and they were offered the opportunity to ask questions.

Next, participants used our QV tool to cast their votes. Voters could allocate their 99 credits across the ten different ballot issues elucidated during the third situated action, to signal the intensity of their conviction per issue. Once all votes were submitted, after 20 minutes, we decimated the voting data and shared the overall results on a central screen. (Voters could also access the results on their devices.) Members could openly ask questions and partake in deliberation about the results.

The new QV tool generated a spectrum of granular data (not only per issue, but comparatively), including how individual responses weigh up to collective results. While most case studies on QV consider only the ranking of issues as ‘the’ result, our tool includes all granular data as well as an analysis of it [13]. In this way, the tool can be used to guide next steps to be taken. For instance, finalizing a broadly supported issue or delaying a contested issue for further deliberation.

3 Results

Our case study yielded a varied set of data, ranging from screenshots taken from our embedded position as a member of the WhatsApp groups, and sketch notes and photos from our first situated action; to a QV tool and ten ballot issues resulting from our second and third situated actions, respectively, and the voting results/data from our fourth action.

Our embedded position as a member of the CLT H-buurt yielded ten screenshots of conversations taking place in the Whatsapp groups. These screenshots documented that one pressing topic about which prospective residents needed to come to a collective decision was the spatial design of the wooncoop dwelling. This

observation informed the four situated actions described above, each yielding different kinds of results.

Our sketch notes and photos of the first situated action indicated that members needed a mechanism to decide collectively about which shared spaces to include and prioritize in a briefing for the project architect. The overall size of the building had been determined which allowed meeting participants to work with a finite resource: square meters.

During meetings, we observed CLT members experimenting with different means of making group decisions. Our photos for instance show CLT members experimenting with online polling tools like Mentimeter and traditional majority (show of hands) voting. Means and tools that at times fell short and failed to accommodate the full complexity of an issue, or failed to include the variety of opinions connected to it. We specifically noticed a general uneasiness with traditional A/B voting, as it did not satisfactorily include the diverse and nuanced opinions of all members. Particular issues might affect certain members or their situation disproportionately. We observed how the results of A/B votes were questioned, instead of producing the anticipated decision, and thus undermining the method. Re-opening democratically decided upon issues elicited mistrust in the decision-making process. Our notes show that, during these meetings, CLT members expressed that they were wary of how particular members' agendas could influence critical decisions that affect the entire group. For instance, members who had been with the CLT longer were concerned that more recent members could influence certain issues, simply because they had a 'louder voice.' At the same time, there was an outspoken desire to include all members' preferences according to the democratically stated mission of the CLT.

Our sketch notes also document a moment at which the briefing was stalled, as different residents had expressed different desires or needs and could not reach consensus about how to spatially design these in their dwelling. Taken together, these in situ observations drove our decision to introduce QV software as an alternative means of collective decision making.

Our second situated action yielded a QV tool for collective decision making regarding the spatial design of the wooncoop [12]. When setting up a new ballot using our tool, the host can input and sequence issues and determine the number of credits each voter receives. The main screen includes a numbered overview of all ballot issues, a credit-to-vote legend indicating the (quadratic) cost of each vote, and a real time overview of how the user is voting. Voters can allocate credits from their pool of credits over all the issues—either in favor or opposed. Credits can be reallocated if the voter changes their mind or reset in their totality. The credits have been conceptualized as opaque tokens, as tokens were considered a familiar form of spending by participants. This design was loosely based on the interactive article mentioned above [3]. To use the tool, individual voters are sent a sign-in link for the voting round and can choose to remain anonymous or share their identity with the group. Voting can be decentralized, if need be.

When a ballot is submitted, the voter is linked to the results page which includes an overview of their voting results juxtaposed with the overall group results. Here, one can delve deeper into the analytics. For instance, one can see if a particular issue was hotly contested, or did or did not receive a lot of traction. To help make

more sense of this analytical information we designed 'badges' to indicate, for instance, which issues received the most attention (as this might not be the number one ranked issue) and when an issue is polarizing. The design of the badges was driven by shortcomings we observed in the software the CLT was experimenting with and based on our in situ recording of how these tools did not satisfactorily elucidate the diverse and nuanced opinions of all members. The inclusion of these badges means our QV tool can provide in-depth insight into voting data, which could then help the group make different types of decisions: about which issues could be harmoniously taken up, and which ones could merit additional rounds of deliberation.

Our third situated action resulted in 13 pages of notes of 10 interviews and a collection of 33 digital photos of 'shareable' objects. Our notes and photos showed that the ten most frequently mentioned spaces, and/or space related issues, for the wooncoop were:

- Space for collective childcare;
- shared laundry room;
- shared central kitchen;
- sharing the ground floor with neighbors (non-member CLT members);
- spaces for intergenerational companionship;
- spaces enabling younger residents to care for older residents;
- individual dwellings organized around a shared central space (compound house);
- an open door policy;
- using sustainable materials even if more expensive; and
- green spaces (for instance, gardens).

These ten spaces and space-related issues were converted into ballot issues for the quadratic voting round, to decide on the spatial design of the wooncoop. See Figure 2 for an example.

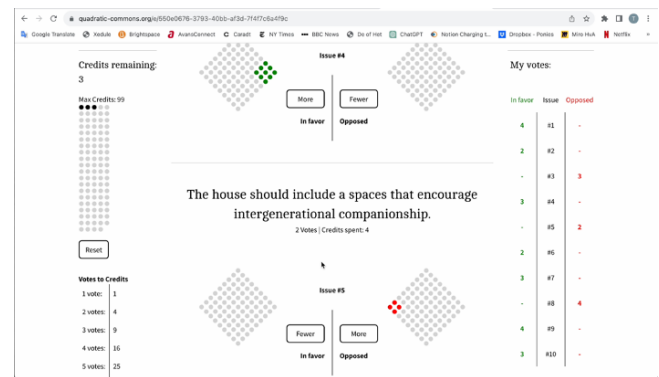


Figure 2: Screenshot of credits being allocated using our QV voting tool, highlighting one of the ten ballot issues.

Our fourth situated action yielded voting results consisting of the data reflecting all individual votes, as well as the group voting result. It also yielded an analysis of this data using the badges mentioned above to mark these instances [13] to help the group make decisions about which issues could be harmoniously taken up, and which ones could merit additional rounds of deliberation.

Remarkably, during a session moderated by the researchers following the vote, wooncoop members chose to exclusively view and deliberate the group voting result, as opposed to scrutinizing individual responses. Additionally, we noted that each voter used all 99 credits, thus exerting all their individual voting power. One participant pointed out, ‘If I don’t utilise them, it would just be a waste.’ Participants even redistributed their credits, thus reconfiguring their initial spread of votes across the ballot, to be able to spend all their credits. This reassigning of credits indicates that individual preferences are paramount during voting. But in contrast, when the group result was revealed, it was ubiquitously accepted without any discord.

These results were not merely a ranking of the ‘winning’ issues (most in favor votes), but an assemblage of granular data per ballot issues. This is because credits can be spent according to conviction: more/less in favor of or more/less opposed to each issue. Hence, the amount to which each voter is more in favor of or opposed to each issue is reflected in the voting data. Our tool comparatively analyzed and sorted this spectrum of data to determine, for instance, which ballot issues attracted the most attention (most net votes), and which ones were polarizing. Because this was our initial attempt to sort a plethora of QV generated data, we chose pronounced indicators (most net votes and polarisation) and chose to include remaining data in a drop-down menu under each issue, so as not to overwhelm participants.

The ballot issue ‘space for collective childcare’ attracted the most attention, indicated by the badges we designed, See Figure 3. The issue received the most votes in favor of, and no opposing votes. This data clearly pointed to the supreme importance of this issue. It was imperatively taken up to be communicated to the project architect in the briefing.

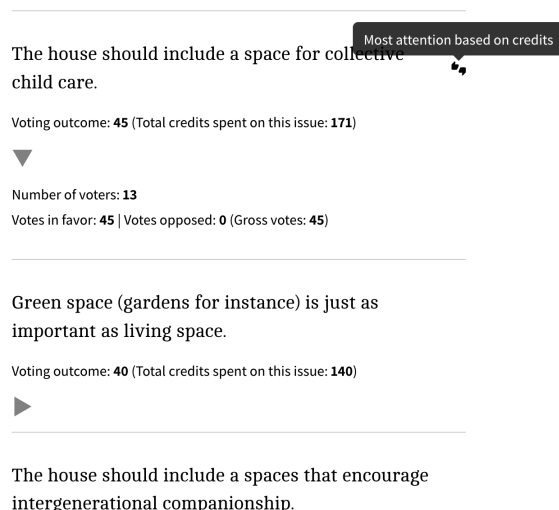


Figure 3: detail of the results screen, showing votes and credit spend on each issue, as well one of our badges.

The most polarizing issue was observed to be ‘sharing the ground floor with neighbors (non-member CLT members)’. The vote on this was split down the middle, as indicated by the badges we

designed. This led to pointed and focused deliberation during an immediate follow-up session about this particular issue moderated by the researchers. During the sessions, one participant remarked, ‘I don’t think it [sharing the ground floor with] will give us the full benefit of being community. We’re going to be isolated because we want to protect our territory.’ In response to which another participant pointed out, ‘Maybe we have a space that is only for playing on the ground floor. . . it also depends on how the place is designed.’ The participants were clearly not yet ready to decide.

The data also displayed which issues garnered attention and no opposition, the so-called ‘no-brainers’ like including green spaces.

By badging these results our tool helped to whittle down the ballot, by indicating which issues merit more deliberation, and which ones the group could more readily harmoniously decide upon— thus supporting different courses of action. For instance, a clearly polarizing issue was put up for more deliberation, while a ‘no brainer’ could be immediately taken up.

4 Discussion

Our case study results indicate potential of QV for cohousing communities by supporting that cohousing is a domain that is ripe for QV in the following two ways.

Firstly, our case study supports this by illustrating that communities need to make many different decisions together. Before a cohousing community like CLT H-buurt can even begin to make decisions like the allocation of space, it needs to first determine together the issues that require decision making, as well as the mechanism through which those decisions are to be collectively made. Secondly, our results support that issues to be decided on collectively affect members differently. For instance, some participants cared more about green spaces as opposed to allocated space for intergenerational companionship and thus casted their votes accordingly.

Besides supporting the potential of QV for cohousing communities, our case study results also reveal three important insights for the operationalization of QV in this domain.

The first insight for the operationalization of QV in cohousing communities is that elucidating ballot issues in situ can improve the deliberation process. The results from our first situated action illustrate that the vast number of group decisions cohousing communities must take may cause some to get bogged down in discussions, which can thwart progress during an already particularly laborious process. The results from our third action shows that elucidating ballot issues in situ, i.e. literally making these kinds of discussions vote-able, can improve deliberation processes by concretizing issues. Combined, these results indicate that CLT members assume a more serious decision-making posture when encountering a formally prepared ballot, in comparison to the informal and long-winded discussions we initially observed from our embedded position.

The second insight for the operationalization of QV resulted from our fourth situated action, which indicates that QV tools designed from embedded positions can streamline the deliberation process. Applying the granular data per ballot issue allowed us to indicate which issues drew the most attention (most net votes), such as the space for collective childcare, and which issues were

polarizing, such as the issue of whether to share (access to) the ground floor of the wooncoop with (non-member CLT member) neighbors. By ‘badging’ these results, our tool helped to whittle down the ballot, and shift more deliberation time to polarizing issues, while a ‘no brainer’ could be immediately taken up. This in-depth analysis and application of granular data by our tool differs from how QV results have been designed to be displayed and shared in the legislative context. In the legislative context, QV results are a singular ranking (most in favor votes), which does not enable underlying and nuanced data to be considered during deliberative processes. This finding suggests that QV can improve collective decision-making processes by streamlining the deliberation process, by designing software tools to behave in correspondence to the needs of the decision-making process.

This does raise the question of how exactly to sort the resulting data in the interest of a particular community. In other words, when does the community describe a result as polarizing, or at exactly what point is there enough commonality about a certain issue to make a binding decision? Our case study results suggest a possible future improvement could be determining and working with thresholds, defined together with the community, for assigning badges to delineate particular group results.

The third insight for the operationalization of QV is that it can make deliberation more harmonious and reflective, given that a QV tool displays both individual as well as group results and provides the option to anonymize user identities. The deliberation directly following the QV vote during our fourth situated action was a productive reflection on the votes rather than the ostracizing of specific individuals (‘finger pointing’) we observed in the first action. Also, the collective outcome of the fourth situated action was ubiquitously accepted without any discord, which contrasted with the re-opening of issues observed in the first action.

This points to a possible limitation for QV in the context of cohousing. Since collective decision making by cohousing communities typically takes place over a long period of time, it can be difficult to determine at which point a decision becomes binding. In other words, when is QV a tool for deliberation and when is QV a tool for making a final decision?

5 Conclusion

Based on the two ways in which our case study shows that the domain of cohousing is ripe for QV discussed above, we conclude that our situated design case study supports our conjecture that QV holds potential for decision making in cohousing communities. Furthermore, to operationalize QV in this domain, based on our case study results, we recommend that ballot issues are elucidated in situ and that QV software tools are designed from embedded positions, as well as that they display both individual as well as group results and provide the option to anonymize user identities.

Taken together, this ripeness of the domain and these recommendations for operationalization, encourage further research on the potential of QV, in the context of cohousing as well as in other domains in which many decisions need to be made collectively on issues that typically affect individual members differently. Moreover, our case study brought to the forefront that such research

should have attention for decision-making processes, as opposed to merely evaluating a singular voting result.

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